Fiscal Year:	FY 2021	**Task Last Updated:**	FY 01/21/2022
PI Name:	Sargsyan, Ashot M.D.		
Project Title:	Multipurpose Ultra-Portable Ultrasound Imager with Artificial Intelligence (AI) Capability		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	asargsyan@wylehou.com	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	
Organization Name:	KBR/NASA Johnson Space Center		
PI Address 1:	2400 NASA Pkwy Mail Code PMO		
PI Address 2:			
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	36
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	TRISH--Focused Investigations
Start Date:	06/01/2018	**End Date:**	09/30/2023
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 9/30/2023 per E. Urquieta/TRISH (Ed., 10/20/22) NOTE: End date changed to 9/30/2022 per E. Urquieta/TRISH (Ed., 8/3/21) NOTE: End date changed to 9/30/2021 per E. Urquieta/TRISH (Ed., 4/14/21) NOTE: End date changed to 02/28/2021 per TRISH (Ed., 8/20/20) NOTE: End date changed to 7/31/2019 per E. Urquieta/TRISH (Ed., 5/23/19)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Ebert, Douglas Ph.D. (KBR/NASA Johnson Space Center) Byrne, Vicky M.S. (KBR/NASA Johnson Space Center) Shah, Ronek D.O. (NASA Johnson Space Center) Pattarini, James M.D. (NASA Johnson Space Center)		
Grant/Contract No.:	NNX16AO69A-FIP0007		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Focused Investigation Project The goal of this project is to conduct ground-based and International Space Station (ISS) in-flight evaluation of new solid state ultrasound transducer technology with integrated artificial intelligence (AI) solutions in order to validate their operational usability in continuous microgravity and their innovation potential for operational medical support across all phases of ISS missions. This technology has high potential for valuable contributions to human health and performance capability maturation, in alignment with NASA's capability-based development goals. The investigation team will: 1) Select a high-Technology Readiness Level (high-TRL), Digital Imaging and Communications in Medicine (DICOM) compliant commercial off-the-shelf (COTS) ultrasound system, which incorporates new single-probe imaging technology, advanced data management with built-in AI solutions, intuitive user interface, and connectivity. The Butterfly iQ system (Butterfly Network Inc., N.Y., N.Y.), which is newly Food and Drug Administration-cleared (FDA-cleared) and will become available in 2018, is the only device on the market to meet all the above criteria. Most of its features are not available with other devices in the market. 2) Acquire iQ (2 units), other necessary equipment, and supplies for ground-based evaluation (Phase 1) 3) Perform a ground-based evaluation of iQ (Phase 1) a. Review hardware and electronics, interfaces, and operating functionality through analysis of manufacturer-provided specifications and testing data; b. Test the iQ (all units) using an approved quality assurance phantom, in parallel with a full-featured device for comparison purposes (GE VividQ or similar, i.e. ISS Ultrasound 2), including assessment of: contrast resolution at multiple depths; axial, lateral, and elevational resolution; penetration depth; c. Perform standardized multi-target imaging tests in human subjects by two trained sonography specialists (Ultrasound small and medium-sized enterprises (SMEs) in ten (10) human subjects (human-in-the-loop engineering evaluation with appropriate NASA approval). These tests will be duplicated with a full-featured system (similar to the current ISS Ultrasound 2) for comparison; Testing will be used to assess: • general usability and human factors, • image fidelity and limitations at three different depth ranges, • comparison of iQ with a full-featured system similar or identical to ISS Ultrasound 2. d. Perform standardized imaging tests and comparison with full-featured portable systems, by point of care ultrasound (POCUS) SME physician(s) using point-of-care ultrasound (POCUS) protocols (focused assessment with sonography in trauma (FAST) exam and lung ultrasound protocol) in 5 (five) human subjects (human-in-the-loop engineering evaluation, with appropriate NASA approval), and ultrasound-guided vascular access using a COTS ultrasound-guided vascular access phantom. e. Assess effectiveness of remote guidance using image streaming if available at the time of this evaluation. NDA and additional coordination with the Manufacturer for access to pre-release software may be required. If feasible, this will be performed by using novice operators in a limited scanning test in 5 (five) human subjects (human-in-the-loop engineering evaluation). f. Test effectiveness of AI solutions provided by the Manufacturer with novice operators using 10 human subjects (human-in-the-loop engineering evaluation) and observed by Ultrasound SMEs. 4) Perform analysis of formalized ratings, comments, and observations from all assessments and imaging tests (3 a-f above) by ultrasound SMEs with determination of appropriateness of iQ as a prototype for NASA human health and performance capability. 5) Prepare a Ground-Based Evaluation (Phase 1) Report. Presentation of results to TRISH and appropriate NASA organizations/boards.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This investigation examines multiple facets of novel ultrasound imaging technology, as well as its closer integration with delivery of medical care in various mission profiles and mission phases of human spaceflight. Scientific approach to device and technology evaluation, which includes comprehensive study of technical features, usability, data quality, human factors, and roles in various aspects of medical care, will result in well-substantiated recommendations. Multiple features of the device represent significant innovation. Of those, the single-probe solution based on Capacitive Micromachined Ultrasound Transducer (CMUT) with thousands of elements integrated with digital electronics in a highly compact design is highly attractive for a highly constrained spaceflight. Further, integration of deep-learning-based tools in the device software are a major step in the right direction of increasing crew autonomy and automated data processing. These and other features are closely examined by the team and will take a prominent place in the final report of the investigation. At this time, data analysis has not been completed, hence the information for this section remains limited and hypothetical. The goal of this project is to conduct ground-based and ISS in-flight evaluation of new ultrasound transducer ("ultrasound on a chip") technology coupled with artificial intelligence (AI) solutions to validate their operational usability in astronaut care including flight, and their innovation potential for operational medical support across all phases of ISS missions. The novel ultra-portable imaging device prototype represents significant innovation for both short- and long-term needs of operational space medicine. The previously unavailable transducer technology allows using one probe for all imaging applications. Importantly, the device is based on philosophy of continued development, expansion of "smart" features, novice operator- friendly features, use of a cloud-based resources with upgradeability through software and firmware evolution. The device is DICOM-compliant and FDA-approved. Thus, the potential benefits identified by this proposal are: • Concept of operations with near-instantaneous availability of imaging technology, connectivity, and streamlined data handling; • Ultra-portability and small footprint (mass, power, volume) with single-probe solution for all imaging needs; • Deep learning-based tools with automatic target recognition and operator assistance; upgradeability of firmware and software; • Preset-based, intuitive operation supporting non-medical operators, yet offering multiple imaging applications pertinent to space medicine; • Resource savings. The investigation team is in the process of analyzing data. According to preliminary review of data, this device can serve

Task Progress:	as an excellent basis for further development and optimization. The approach entirely aligns with the concept of developing autonomous healthcare capabilities in conditions of exploration flight under conditions of multiple constraints, including absent real-time communications with Earth. In the original proposal the investigation team has proposed to: 1) Select a high-TRL, DICOM-compliant commercial off-the-shelf (COTS) ultrasound system, with a new single-probe imaging technology, advanced data management with upgradeable deep learning-based solutions, intuitive user interface, and connectivity. The Butterfly iQ (iQ) system (Butterfly Network Inc., N.Y., N.Y.), which has been FDA-cleared and available since 2018, is the only device on the market to meet the above criteria. Most of its features are not available in any single device in the market; 2) Acquire iQ (2 units), other necessary equipment, and supplies for ground-based evaluation (Phase 1); 3) Perform a ground-based evaluation of iQ (Phase 1); a. Review hardware and electronics, interfaces, and operating functionality through analysis of manufacturer-provided specifications and testing data; b. Test the iQ (all units) using an approved quality assurance phantom, in parallel with a full-featured device for comparison purposes (GE VividQ or similar, i.e. ISS Ultrasound 2), including assessment of: contrast resolution at multiple depths; axial, lateral, and elevational resolution; penetration depth, and others features; c. Perform standardized multi-target imaging tests in human subjects by two trained sonography specialists (Ultrasound SMEs) in ten (10) human subjects (human-in-the-loop engineering evaluation with appropriate NASA IRB approval). These tests will be duplicated with a full-featured system (similar to the current ISS Ultrasound 2) for comparison; Testing will be used to assess: • general usability and human factors; • image fidelity and limitations at three different depth ranges; • comparison of iQ with a full-featured system similar or identical to ISS Ultrasound 2. d. Perform standardized imaging tests and comparison with full-featured portable systems, by POCUS SME physician(s) using point-of-care ultrasound (POCUS) protocols (FAST exam and lung ultrasound protocol) in 5 (five) human subjects (human-in-the-loop engineering evaluation, with appropriate NASA approval), and ultrasound-guided vascular access using a COTS ultrasound-guided vascular access phantom; e. Assess effectiveness of remote guidance using image streaming if available at the time of this evaluation. NDA and additional coordination with the Manufacturer for access to pre-release software may be required. If feasible, this will be performed by using novice operators in a limited scanning test in 5 (five) human subjects (human-in-the-loop engineering evaluation); f. Test effectiveness of AI solutions with novice operators using 10 human subjects (human-in-the-loop engineering evaluation) and observed by Ultrasound SMEs. 4) Develop the analysis framework and perform analysis of imagery, formalized ratings, comments, and observations from all assessments and imaging tests (3 a-f above) by ultrasound SMEs with determination of appropriateness of iQ as a prototype for NASA human health and performance capability; 5) Prepare a Ground-Based Evaluation (Phase 1) Report. Presentation of results to TRISH and appropriate NASA organizations/boards. In addition to the original goals and objectives, the team has initiated and advanced in the preparations for a technology demonstration of the iQ device using ISS as a technology development testbed. This activity is approved through a Supplement to the proposal, which sets a goal to demonstrate functionality and usability of an ultra-portable ultrasound system (Butterfly iQ, Butterfly Network, USA) aboard the International Space Station in a technology demonstration. At the time of this interim report, all data collection activities have been completed, and the team has started developing the data analysis framework and outlining the Final Report. Early findings of the investigation, which appear to be favorable, have been shared with the aerospace medicine community through interim reports to TRISH, as well as through presentations at appropriate NASA and public forums.
Bibliography Type:	Description: (Last Updated: 05/19/2026)
Abstracts for Journals and Proceedings	Sargsyan AE, Douglas E, Everson A, Suri K, Cole R, Pattarini JM, Scheuring RA, Uhl SA, Byrne V, Shah R. "Evaluation of novel ultrasound imaging technology from the perspective of space medicine." 2020 NASA Human Research Program Investigators' Workshop Galveston, TX, January 27-30, 2020. Abstracts. 2020 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 27-30, 2020. , Jan-2020
Abstracts for Journals and Proceedings	Sargsyan AE, Ebert D, Everson A, Suri K, Cole R, Pattarini JM, Scheuring RA, Uhl SA, Byrne V, Shah R. "Evaluation of the Butterfly iQ ultrasound imager for use in human spaceflight programs." 2021 NASA Human Research Program Investigators' Workshop, Virtual, February 1-4, 2021. 2021 NASA Human Research Program Investigators' Workshop, Virtual, February 1-4, 2021. , Feb-2021